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The Editors-in-Chief
Machine Learning
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Dear Editors,

Please consider the enclosed manuscript, "*VOUCH: Anytime-Valid Certificates for Machine Unlearning via Paired-Canary Hypothesis-Betting*," for publication in *Machine Learning*. In keeping with double-blind review, the manuscript itself carries no author information; I am the sole author, and my identity and affiliation appear only in this letter.

The paper asks a question that regulation has made urgent and that current practice cannot answer: when a provider runs an unlearning algorithm to honour a deletion request, can anyone certify—with controlled error—that forgetting took place? Existing evaluations are fixed-sample tests that are unsound as membership evidence, void their guarantees under the continuous monitoring real audits involve, and lean on retrained reference models that are unaffordable at scale. VOUCH removes all three obstacles at once: *paired ghost canaries*, of which a fair coin admits one twin into training and later into the forget set, make the within-pair score difference symmetric under exact unlearning—an exact, distribution-free null over which betting e-processes deliver an anytime-valid confidence sequence on residual membership advantage, an (ε, α)-forgetting certificate, and a revocation alarm. A central finding is that certificates and alarms must compose across deletion waves in opposite directions—consensus versus product—and that the intuitive alternatives falsely certify up to 99.6% of the time. The framework is validated by a two-thousand-seed calibration study under adversarial optional stopping, certification of four unlearning methods on the TOFU and MUSE-News benchmarks across several architectures, and recoverability probes, at a verifier cost two orders of magnitude below shadow-model evaluation. As stated in the manuscript's data availability statement, all code, per-seed results, and the execution harness are publicly released; the repository reference is withheld from the blinded manuscript and is:

`github.com/vinhqdang/VOUCH-Verifiable-Online-Unlearning-Certification-via-Hypothesis-betting`

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I have no conflicts of interest to declare. I believe the combination of game-theoretic statistics with a verifier-generated null, and the resulting first anytime-valid certification framework for machine unlearning, fits *Machine Learning*'s scope well, and I hope you will find it suitable for review.

Thank you for your consideration.

Sincerely,

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